

SMS in हिन्दी

English, spoken only by a minority in India, is ironically the language of choice in computers, mobile phones and almost any other gadget. But soon this will change. A Canadian firm called Zi Corporation is working to let you SMS in a local language, and Hindi is its first target. But is it so hard to provide an option to send SMS in any vernacular language?

Mobile operators currently offer a pretty mundane vernacular SMS service. A user has to select the necessary option, and use the English alphabet to enter a message (which is a phonetic representation of a regional language). This message is then sent to an operator, who converts it into the correct font, and sends it back to the user. Now, the user can send it to the intended receiver. This is a long procedure, and the entire exercise is like typing a *desi* word in a chat window, coupled with the latency of waiting to hear from the operator.

The other option, available in a few phone models, such as the Nokia 3350, is akin to the method of installing a regional language font and entering text, using a phonetic keyboard. Mildly put, this is a big hassle. First, one has to change the language to Hindi; then enter the Messages menu. Typing a message in Hindi is rather cumbersome without the predictive mode. As a result, it is quite key-press intensive and monotonous, as there are up to nine Hindi letters and symbols on one key.

We tried sending a Hindi message from the Nokia 3350. It was received and displayed perfectly on a Nokia 3315, without switching the phone-language to Hindi. However, the message could not be edited easily on the 3315. Next, we forwarded the message to a



Nokia 3310. The 3310, which does not have a Hindi language pack installed, displayed junk on the screen.

Computers fundamentally deal with combinations of ones and zeroes. Unicode provides a standard where letters and characters are stored, and understood, by assigning a unique code to each one. Currently, the Unicode standard contains 34,168 unique binary codes for 24 language scripts. However, the Devanagari script, on which Hindi is based, is not fully supported by Unicode.

The incompatibility with

Unicode is largely attributed to ligatures in the Devanagari script. A ligature is a character, letter or type, such as æ, which is obtained by combining two or more letters. There are almost 400 ligature combinations found in the Devanagari script, compared to a handful in English. The other problem posed by ligatures in Hindi, is that the resulting combination can be totally different in shape from the initial characters used—the 'ksha' letter is a common ligature.

The scenario can become pretty overwhelming, because of the problems and technology involved.

Zi Corporation plans to introduce the first predictive text input for mobile phones

क् + ष = क्ष

in Hindi. eZiText Hindi uses an innovative keyboard layout, but the predictive text input is similar to that used in phones today; multiple key taps are not necessary for character selection.

A flash demo on their Web site, www.zicorp.com, demonstrates how easy it is to type messages using eZiText—the word dhanyavad can be typed in six key-presses. Even on a desktop QWERTY keyboard, using the free MyHindi font, it takes more than six key-presses.

Mobile phones in the Indian market today typically do not handle ligatures, and the few that do are incomplete or contain ligature display errors. To address these issues, Zi provides a package of ligature rules

to help manufacturers localise their device implementations, by resolving display and other interface issues.

eZiText uses the same core engine for all alphabetical languages. Accordingly, most eZiText implementations have at least two languages, generally a regional language, say Hindi, and English. As a result, language input in the same message can also be mixed.

Native speakers of other Indian languages need not feel left out. Zi also offers another product—eZiTap, a predictive multi-tap system, which is language independent. eZiTap has a predictive engine, but no word database. The engine 'learns' words, which are multi-tapped the first time, and predicts them in the future. This offering has the potential to take SMS to every dialect spoken in India. However, one must note that a handset needs to support the font and script of the language in question.

But, do you require another mobile phone to start messaging in your mother tongue? Gary Mendel, VP Sales & Marketing, told us, "Several licensees are currently implementing eZiText Hindi for multiple products due in 2003. Zi has over 70 customers, including Sony Ericsson, Alcatel, Samsung, LGE, Kyocera and DBTel."

Hindi SMS should be a reality very soon, and it should set the ball rolling for full-fledged messaging in other regional languages too. Here's hoping that *Schwaag ki Maa* will be able to message her son in any language she pleases. ■